

# KIT101 Assignment2

## Coffee Shop Analytics Dashboard

[coffee-shop-sales-revenue.csv.zip](#)

### 1. Project title, user, and problem

#### Title

Coffee Shop Sales Analytics Dashboard

#### Intended User

A coffee shop manager who wants to understand sales performance and identify popular products.

#### Problem Statement

The coffee shop collects sales data every day, but it is difficult for managers to quickly identify trends, top selling products, and revenue patterns. This program will provide a simple analytics dashboard that loads sales records, analyses performance and generates a sales report.

### 2. Scope

#### Core Features I Will Complete

1. Load coffee shop sales data records from a CSV file.
2. Validate and clean the data. Make sure there are no missing values or invalid quantities.
3. Calculate for the sales reports and see the product performance, including total revenue, daily revenue, best selling products, and rankings.
4. Create a sales report with visual charts.
5. Business Recommendation summary report based on the analysis.

### 3. Program structure

```
main.py
├── main()
│   ├── display_founction()
│   ├── load_data()
│   │   ├── data_handler.py
│   │   │   ├── load_csv()
│   │   │   └── validate_data()
│   ├── calculate_revenue()
│   │   ├── analytics.py
│   │   │   ├── day_total_revenue()
│   │   │   ├── month_total_revenue()
│   │   │   └── revenue_by_category()
│   ├── analyse_products()
│   │   ├── analytics.py
│   │   │   ├── top_selling_products()
│   │   │   └── lowest_selling_products()
│   ├── generate_charts()
│   │   ├── visualisation.py
│   │   │   ├── revenue_trend_chart()
│   │   │   └── category_sales_chart()
│   └── save_report()
```

### 4. Important data and files

The main data structure will be a list of dictionaries. Data structure for a sale:

```
{
  "date": "2026-08-01",
  "item": "Latte",
  "quantity": 2,
  "price": 5.50
}
```

The program report will save as `SaveReport.txt`.

## 5. Five representative planned tests.

| Function            | Input             | Expected Result            | Why it matters |
|---------------------|-------------------|----------------------------|----------------|
| load_csv()          | Valid CSV         | Data loaded correctly      | Normal case    |
| load_csv()          | Missing file      | Error message              | Error case     |
| validate_data()     | Negative quantity | Record rejected            | Invalid data   |
| calculate_revenue() | Known sample data | Correct revenue calculated | Accuracy       |
| top_products()      | Sample dataset    | Correct ranking returned   | logic          |

## 6. Risks and responses

| Risk            | How I Will Reduce It                   |
|-----------------|----------------------------------------|
| Missing value   | Do data cleansing before program start |
| Not enough time | Try to follow the planning time        |

## 7. Development plan

| Week    | Planned Work                                              | Evidence                        |
|---------|-----------------------------------------------------------|---------------------------------|
| Week 10 | Select dataset, implement data loading and data cleansing | CSV loads successfully          |
| Week 11 | Implement analytics functions and testing                 | Revenue calculations pass tests |
| Week 12 | Create charts and recommendations                         | Dashboard generates reports     |
| Week 13 | Complete report, testing and presentation                 | All tests pass                  |

## 8. GenAI decision

This example does not propose an AI-assisted module. If I later proposed using GenAI for one self-contained module, I would first discuss the scope change with the tutor and record the required information in `GENAI_DISCLOSURE.md`. I would still write my own `test_` functions for that module and explain how it works.